

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0843

Pinnacle Peak Aps 230 kV
August 03, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0843. The Interconnection Customer proposes a 132.7 MW solar photovoltaic facility interconnecting at the Pinnacle Peak Aps 230 kV substation in the Ohio area. The request was received on August 03, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 379 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,587,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
STEVENS RD - CONESVILLE STA 230kV Line	CONESVILLE STATION - MUSKINGUM RIV 500kV Line	128.9	516	5.5
BARRETT - MINCO SUBSTATION 230kV Line	BARRETT - NORWICH 500kV line	108.9	620	11.3
CONESVILLE STA - SUNNYVALE 230kV	DRESDEN - CONESVILLE STATION 500kV line	107.1	566	5.1
SUNNYVALE SUBSTATION - DRESDEN 230kV	DRESDEN - CONESVILL STA 500kV	105.5	602	5.1
TALENENERGY MARTINS CRK LLC WILLIAMSPORT 500/345 kV Transformer Bank	HILLTOP 500-345 kV Transformer	101.3	2,280	5.9
CALDWELL - PINEY CRK PROJECT 345kV L	HILLTOP - PINEY CRKPROJECT 500kV L	100.6	2,276	7.9
BLACKROCK - SHAWVILLE STATION 345kV L	CARBON CTR 500/345kV Transformer Bank	100.6	2,111	7.0
TALENENERGY MARTINS CREEK LLC HARRISBURG - LITTLE CHUTE 345kV Line	CORONET - FORT HUMBUG 500kV	100.3	2,249	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
STEVENS RD - CONESVILLE STATION 230kV	Rebuild line with higher-capacity conductor	\$40,116,000	2% (\$850,000)
MINCO - BARRETT 230kV	Reconductor line	\$4,150,000	5% (\$199,000)
CONESVILLE STA - SUNNYVALE 230kV	Reconductor line	\$10,286,000	2% (\$254,000)
SUNNYVALE - DRSDEN 230kV Line	Reconductor line	\$3,816,000	2% (\$94,000)
TALENENERGY MARTINS CREEK LLC WILLIAMSPORT 500/345kV Autotransformer	Replace transformer with higher-rated unit	\$26,835,000	1% (\$352,000)
PINEY CREEK PROJECT - CALDWELL 345kV	Reconductor line	\$34,190,000	1% (\$441,000)
BLACKROCK - SHAWVILLE STATION 345kV L	Reconductor line	\$23,714,000	1% (\$309,000)
LITTLE CHUTE SUBSTATION - TALENENERGY MARTINS CREEK LLC HARRISBURG 345kV L	Reconductor line	\$6,696,000	1% (\$88,000)
	Total allocated to project		\$2,587,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.